Chapter: Light – Reflection and Refraction

1. Which one of the following materials cannot be used to make a lens?
    a. Water
    b. Glass
    c. Plastic
    d. Clay
2. The image formed by a concave mirror is observed to be virtual, erect and larger than the object. Where should be the position of the object?
    a. Between the principal focus and the centre of curvature
    b. At the centre of curvature
    c. Beyond the centre of curvature
    d. Between the pole of the mirror and its principal focus.
3. Where should an object be placed in front of a convex lens to get a real image of the size of the object?
    a. At the principal focus of the lens
    b. At twice the focal length
    c. At infinity
    d. Between the optical centre of the lens and its principal focus.
4. A spherical mirror and a thin spherical lens have each a focal length of –15 cm. The mirror and the lens are likely to be
    a. both concave.
    b. both convex.
    c. the mirror is concave and the lens is convex.
    d. the mirror is convex, but the lens is concave.
5. No matter how far you stand from a mirror, your image appears erect. The mirror is likely to be
    a. only plane.
    b. only concave.
    c. only convex.
    d. either plane or convex.
6. Which of the following lenses would you prefer to use while reading small letters found in a dictionary?
    a. A convex lens of focal length 50 cm.
    b. A concave lens of focal length 50 cm.
    c. A convex lens of focal length 5 cm.
    d. A concave lens of focal length 5 cm.
7. We wish to obtain an erect image of an object, using a concave mirror of focal length 15 cm. What should be the range of distance of the object from the mirror? What is the nature of the image? Is the image larger or smaller than the object? Draw a ray diagram to show the image formation in this case.
8. Name the type of mirror used in the following situations.
    a. Headlights of a car.
    b. Side/rear-view mirror of a vehicle.
    c. Solar furnace.

    Support your answer with reason.

9. One-half of a convex lens is covered with a black paper. Will this lens produce a complete image of the object? Verify your answer experimentally. Explain your observations.
10. An object 5 cm in length is held 25 cm away from a converging lens of focal length 10 cm. Draw the ray diagram and find the position, size and the nature of the image formed.
11. A concave lens of focal length 15 cm forms an image 10 cm from the lens. How far is the object placed from the lens? Draw the ray diagram.
12. An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.
13. The magnification produced by a plane mirror is +1. What does this mean?
14. An object 5.0 cm in length is placed at a distance of 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position of the image, its nature and size.
15. An object of size 7.0 cm is placed at 27 cm in front of a concave mirror of focal length 18 cm. At what distance from the mirror should a screen be placed, so that a sharp focussed image can be obtained? Find the size and the nature of the image.
16. Find the focal length of a lens of power – 2.0 D. What type of lens is this?
17. A doctor has prescribed a corrective lens of power +1.5 D. Find the focal length of the lens. Is the prescribed lens diverging or converging?

Chapter: Electricity

1. A piece of wire of resistance R is cut into five equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R′, then the ratio R/R′ is –
   a. 1/25
   b. 1/5
   c. 5
   d. 25
2. Which of the following terms does not represent electrical power in a circuit?
   a. I^2R
   b. IR^2
   c. VI
   d. V^2/R
3. An electric bulb is rated 220 V and 100 W. When it is operated on 110 V, the power consumed will be –
   a. 100 W
   b. 75 W
   c. 50 W
   d. 25 W
4. Two conducting wires of the same material and of equal lengths and equal diameters are first connected in series and then parallel in a circuit across the same potential difference. The ratio of heat produced in series and parallel combinations would be –
   a. 1:2
   b. 2:1
   c. 1:4
   d. 4:1

5. How is a voltmeter connected in the circuit to measure the potential difference between two points?
6. A copper wire has diameter 0.5 mm and resistivity of 1.6 × 10–8 Ω m. What will be the length of this wire to make its resistance 10 Ω? How much does the resistance change if the diameter is doubled?
7. The values of current I flowing in a given resistor for the corresponding values of potential difference V across the resistor are given below – Plot a graph between V and I and calculate the resistance of that resistor.

I (amperes)	0.5	1.0	2.0	3.0	4.0
V (volts)	1.6	3.4	6.7	10.2	13.2

8. When a 12 V battery is connected across an unknown resistor, there is a current of 2.5 mA in the circuit. Find the value of the resistance of the resistor.
9. A battery of 9 V is connected in series with resistors of 0.2 Ω, 0.3 Ω, 0.4 Ω , 0.5 Ω and 12 Ω, respectively. How much current would flow through the 12 Ω resistor?
10. How many 176 Ω resistors (in parallel) are required to carry 5 A on a 220 V line?
11. Show how you would connect three resistors, each of resistance 6 Ω, so that the combination has a resistance of (i) 9 Ω, (ii) 4 Ω.
12. Several electric bulbs designed to be used on a 220 V electric supply line, are rated 10 W. How many lamps can be connected in parallel with each other across the two wires of 220 V line if the maximum allowable current is 5 A?
13. A hot plate of an electric oven connected to a 220 V line has two resistance coils A and B, each of 24 Ω resistance, which may be used separately, in series, or in parallel. What are the currents in the three cases?
14. Compare the power used in the 2 Ω resistor in each of the following circuits:
    i. a 6 V battery in series with 1 Ω and 2 Ω resistors, and
    ii. a 4 V battery in parallel with 12 Ω and 2 Ω resistors.
15. Two lamps, one rated 100 W at 220 V, and the other 60 W at 220 V, are connected in parallel to electric mains supply. What current is drawn from the line if the supply voltage is 220 V?
16. Which uses more energy, a 250 W TV set in 1 hr, or a 1200 W toaster in 10 minutes?
17. An electric heater of resistance 44 Ω draws 5 A from the service mains for 2 hours. Calculate the rate at which heat is developed in the heater.
18. Explain the following.
    a. Why is the tungsten used almost exclusively for filament of electric lamps?
    b. Why are the conductors of electric heating devices, such as bread-toasters and electric irons, made of an alloy rather than a pure metal?
    c. Why is the series arrangement not used for domestic circuits?
    d. How does the resistance of a wire vary with its area of cross-section?
    e. Why are copper and aluminium wires usually employed for electricity transmission?

Chapter: Magnetic Effects of Electric Current

1. Which of the following correctly describes the magnetic field near a long straight wire?
    a. The field consists of straight lines perpendicular to the wire
    b. The field consists of straight lines parallel to the wire
    c. The field consists of radial lines originating from the wire
    d. The field consists of concentric circles centred on the wire.
2. At the time of short circuit, the current in the circuit
    a. reduces substantially
    b. does not change
    c. increases heavily
    d. vary continuously.
3. State whether the following statements are true or false.
    a. The field at the centre of a long circular coil carrying current will be parallel straight lines.
    b. A wire with a green insulation is usually the live wire of an electric supply.
4. List two methods of producing magnetic fields.
5. When is the force experienced by a current–carrying conductor placed in a magnetic field largest?
6. Imagine that you are sitting in a chamber with your back to one wall. An electron beam, moving horizontally from back wall towards the front wall, is deflected by a strong magnetic field to your right side. What is the direction of magnetic field?
7. State the rule to determine the direction of a
    a. magnetic field produced around a straight conductor-carrying current,
    b. force experienced by a current-carrying straight conductor placed in a magnetic field which is perpendicular to it, and
    c. current induced in a coil due to its rotation in a magnetic field.
8. When does an electric short circuit occur?
9. What is the function of an earth wire? Why is it necessary to earth metallic appliances?